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LESSON PLAN

For

AI Security for Autonomous AI Agents

TGS Ref No: TGS-2025060473

Conducted by

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Version 3.0

Document Version Control Record

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	16 June 2025	Initial release — Core Principles and Ethical Challenges in Generative AI.	Dr Alfred Ang
2.0	17 August 2026	Retitled and rebuilt for generative-AI and autonomous-agent security; five activities and revised assessments added. TSC K/A text unchanged.	Dr Alfred Ang
2.1	18 August 2026	Added prompt-injection practice, PDPA and organisational checklists, human approval, skill/plugin controls and enhanced visuals. TSC K/A text unchanged.	Dr Alfred Ang
3.0	20 August 2026	Rebuilt as a 207-slide evidence-grounded progression from AI history to product boundaries, cases, controls and governance. Activities reordered; unverified claims removed. TSC K/A text unchanged.	Dr Alfred Ang

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1. Course Overview

AI Security for Autonomous AI Agents is a 2 Days · 16 Hours WSQ course that equips learners to identify, analyse and mitigate the security risks introduced by generative AI systems and by autonomous AI agents. The course uses documented cases and clearly labelled realistic simulations, grounded in current security and governance frameworks, including the OWASP Top 10 for LLM Applications (2026), the OWASP Top 10 for Agentic Applications (2026), the NIST AI Risk Management Framework, MITRE ATLAS, the IMDA Model AI Governance Framework for Agentic AI, and Singapore's PDPA and current PDPC guidance. Proposed consultation material is not presented as final law or guidance.

The course covers AI security for BOTH generative AI (prompt injection, data leakage, poisoning, PDPA exposure) and autonomous agents (uncontrolled and destructive execution, tool misuse, agent vulnerabilities and cyber-attack chains).

Skills Framework Alignment

Element	Detail
TSC Title	Generative AI Principles and Applications
TSC Code	ICT-INT-0052-1.1
Proficiency Level	Level 1
Course Code	TGS-2025060473
Duration	2 Days · 16 Hours
Delivery	Instructor-led classroom with facilitated case-study activities

Note: the accredited knowledge and ability statements below are reproduced verbatim. They are taught and assessed as accredited; AI security is the delivery context through which each statement is evidenced.

Learning Outcomes

LO	Learning Outcome
LO1	Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
LO2	Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
LO3	Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

Knowledge Statements (assessed by the Written Assessment)

Code	Knowledge statement
K1	Importance of data quality, preprocessing, model pipeline and model training (e.g., impact of data bias from training data)
K2	Underlying principles, core concepts and theories governing generative AI
K3	Difference between generative and discriminative models
K4	Impact of prompt engineering on the model outputs of generative AI
K5	Generative AI model workings, including training data, algorithms, and outputs

Ability Statements (assessed by the Case Study)

Code	Ability statement
A1	Analyse limitations and potential biases in AI-generated content
A2	Identify the ethical implications and societal impact of AI-generated content
A3	Apply understanding of generative AI principles to use cases
A4	Demonstrate the use of generation AI in diverse applications (e.g., summarisation, inference, reasoning, transformation of content, augmentation of content)
A5	Analyse generative AI models' performance metrics and evaluate the influence of prompt variations

2. Daily Schedule

Each training day runs 9:30am – 6:30pm and delivers 8 instructional hours. A 1-hour lunch break is excluded from instructional time; short tea breaks are counted within the day.

Day 1 — Generative AI Security

Slides 1–98 of the trainer deck.

Time	Topic / Activity	Duration
9:30 – 9:50	Welcome, digital attendance, introductions, learning outcomes and safe-lab boundaries	20 min
9:50 – 10:35	AI history: symbolic systems, machine learning, deep learning, transformers and instruction following (K2)	45 min
10:35 – 10:50	Tea break	15 min
10:50 – 11:35	Generative AI, agentic behaviour and deployed AI agents as overlapping operating layers (K2, K3)	45 min
11:35 – 12:30	Strengths, weaknesses and examples: OpenClaw, Hermes, Claude Code, Codex, ChatGPT Work, DeepSeek model/harness and evidence gates	55 min
12:30 – 1:30	Lunch break	60 min
1:30 – 2:15	Capability, data, token, code-execution, skill/plugin/MCP, network and approval matrices	45 min
2:15 – 3:00	Threat modelling: assets, actors, trust boundaries, authority paths and abuse cases (K1, K5)	45 min
3:00 – 3:45	Activity 1: Threat Modelling a Generative AI Concierge (K2, K3, A4)	45 min
3:45 – 4:00	Tea break	15 min
4:00 – 4:45	Attack surfaces by operating layer: content, state, identity, tools, runtime and network	45 min
4:45 – 5:30	Complete attack chains, source-to-sink reasoning and evidence classification	45 min
5:30 – 6:15	Integrated risk matrix and preview of structural controls, approvals and evidence	45 min
6:15 – 6:30	Day 1 recap, Q&A and PM digital attendance	15 min

Day 1 instructional total: 480 minutes (8 hours), excluding the 1-hour lunch break.

Day 2 — Autonomous Agent Security

Slides 99–207 of the trainer deck.

Time	Topic / Activity	Duration
9:30 – 9:40	Day 1 recap and AM digital attendance	10 min
9:40 – 10:00	Prompt injection, jailbreaks, injection carriers and structural control principles (K4)	20 min
10:00 – 11:00	Activity 2: Prompt Injection and the PDPA Breach Decision (K4, K1)	60 min
11:00 – 11:15	Tea break	15 min
11:15 – 11:40	Personal data, secrets, tokens, retention, transfers and the PDPA breach decision	25 min
11:40 – 12:00	Framework stack followed by detailed agentic-threat and documented-case instruction (A3, K5)	20 min
12:00 – 1:00	Activity 3: Selecting a Security Framework for GenAI and Agents (A3, A5)	60 min
1:00 – 2:00	Lunch break	60 min
2:00 – 2:30	Activity 3 debrief and evidence bridge into the documented incident review	30 min
2:30 – 3:30	Activity 4: Evidence-Based Rogue Agent Incident Review (K5)	60 min
3:30 – 3:45	Tea break	15 min
3:45 – 4:10	Organisational controls: guardrails, HITL, sandbox, egress, monitoring, deployment and incident gates (A1, A2)	25 min
4:10 – 4:35	Activity 5: Agent Governance and the Deployment Gate (A1, A2)	25 min
4:35 – 4:50	Course synthesis, assessment briefing and PM digital attendance	15 min
4:50 – 5:50	Assessment 1: Written Assessment (SAQ) — 5 questions · 40 marks · 60 min	60 min
5:50 – 6:30	Assessment 2: Case Study — 3 questions · 70 marks · 40 min; TRAQOM close	40 min

Day 2 instructional total: 480 minutes (8 hours), excluding the 1-hour lunch break.

3. Activities

Every activity is either a clearly labelled realistic simulation or an evidence-based documented-case review. Each has its own folder under activities/ containing learner instructions, scenarios, discussion questions, checklists and printable PDFs. Full step-by-step facilitation detail is in the Learner Guide.

#	Activity	Day	Duration	K/A assessed
1	Threat Modelling a Generative AI Concierge	1	45 min	K2, K3, A4
2	Prompt Injection and the PDPA Breach Decision	2	60 min	K4, K1
3	Selecting a Security Framework for GenAI and Agents	2	60 min	A3, A5
4	Evidence-Based Rogue Agent Incident Review	2	60 min	K5
5	Agent Governance and the Deployment Gate (capstone)	2	25 min	A1, A2

4. Tools and Resources

Resource	Purpose
Trainer slide deck (PPTX/PDF)	Facilitation, diagrams and case-study prompts
Learner Guide	Detailed step-by-step notes and activity walkthroughs
Activity packs (activities/)	Learner scenarios, discussion questions, checklists and evidence worksheets
LMS/TMS portal	lms-tms.tertiaryinfotech.com — course material and submission
Whiteboard / flip chart	Group threat modelling and kill-chain reconstruction
OWASP LLM & ASI Top 10 (2026)	Reference taxonomies used throughout
IMDA / PDPC publications	Singapore governance and data-protection reference

5. Assessment

Assessment is conducted at the end of Day 2. The briefing for assessment is delivered before the assessment begins.

Instrument	Covers	Detail
Written Assessment (SAQ)	K1 – K5	Written Assessment (SAQ) — 5 short-answer questions, one per K statement
Case Study	A1 – A5 via LO1 – LO3	Case Study — 3 questions, one per Learning Outcome, mapped to the A statements
Format	—	Open Book
Grading	—	Competent / Not Yet Competent
Re-assessment	—	Available for learners assessed Not Yet Competent

Funding eligibility requires a minimum 75% attendance recorded through SSG digital attendance, an assessment outcome of Competent, and completion of the TRAQOM survey.